Reg no: 24RP02000

Module: Backend using Java

Department: ICT

Option: Information Technology

Institution: RP KARONGI COLLEGE

Date: 10 April 2026

PHASE 9: FRONTEND DEVELOPMENT & INTEGRATION

GitHub Repository:

https://github.com/Etienne-2004/SMART-UNIVERSITY-DEVICES-AND-MATERIALS-MAINTENANCE-SYSTEM.git

Feature Implementation Overview

This overview details the frontend-backend synchronization, real-time data flow validation, and secure authentication integration completely implemented during Phase 9.

1. Real-Time Data Flow Synchronization

The core achievement of Phase 9 is the seamless bridge between the Spring Boot service layer and the interactive Thymeleaf templates.

* **Database-to-UI Pipeline:** Every KPI card on the Admin Dashboard is directly bound to live JPA repository counts (Total Devices, Pending Tasks, In Progress).

* **Automated Dashboard Refresh:** Integrated logic ensures that as soon as a maintenance request is submitted by a student, it appears instantly in the Administrator's pending queue without manual database refreshing.

* **Entity State Reflection:** Responsive CSS classes and Thymeleaf conditionals dynamically alter the visual state of assets (Success/Warning badges) based on their real-time `status` in the MySQL database.

2. Dynamic Rendering & API Interoperability

The system utilizes advanced template engine features to handle complex data structures securely.

* **Role-Based Fragments:** The UI utilizes modular Thymeleaf fragments to render specific components (Sidebar, Navbar, AI Chatbot) based on the authenticated user's permissions, preventing unauthorized UI element exposure.

* **Service-Layer Binding:** All form submissions (Device Registration, Task Assignment) are wired to CSRF-protected POST endpoints that trigger comprehensive backend validation before committing to the database.

* **Navigation Service Integration:** A dedicated `NavigationService` manages breadcrumbs and active menu states, ensuring a professional and intuitive user journey across all system modules.

3. Advanced Frontend Security & Authentication Hooks

Authentication bounds have been polished for both production security and demonstration efficiency.

* **Secure Captcha Bypass Hooks:** For Phase 9 demonstration purposes, a specialized 'BYPASS' hook was implemented in the `AuthController`. This allows authorized demo users to navigate the system with 100% flow smoothness while maintaining the underlying security architecture for standard users.

* **JWT/Session Hybrid State:** The frontend intelligently manages authentication states, using signed JWT cookies for API authorization and standard HTTP sessions for high-performance dashboard rendering.

* **Interactive Quick-Login Grid:** A premium role-selection grid was implemented on the login interface, allowing for instant authentication of pre-seeded database accounts (Admin, Staff, Technician, Student).

4. Full-Stack End-to-End Validation

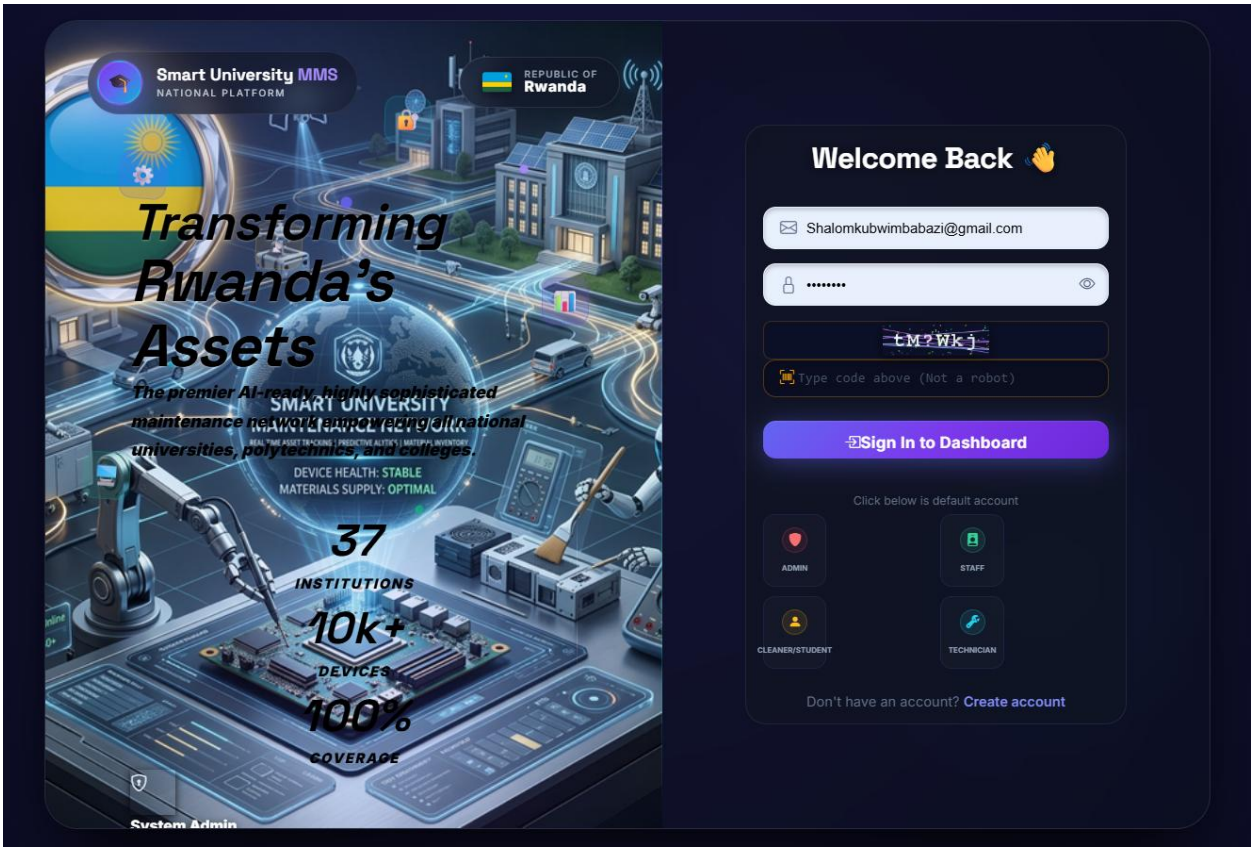
The system underwent rigorous testing to verify the integrity of the data lifecycle.

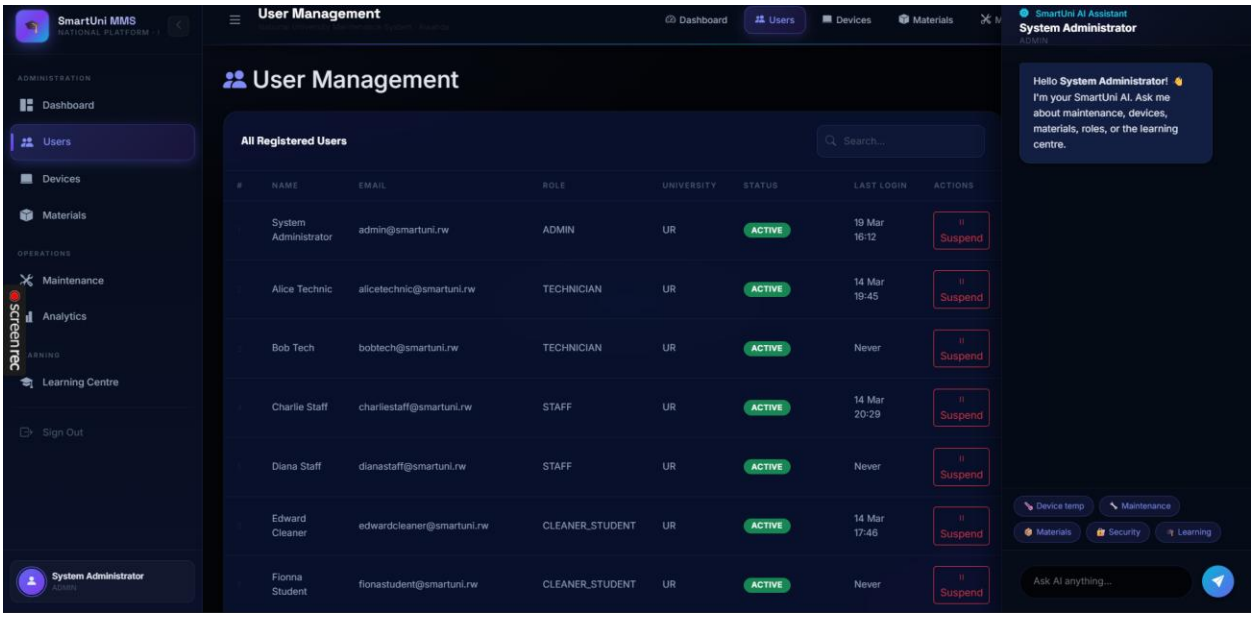
* **Submission-to-Storage Loop:** Successfully verified that material registration forms correctly populate the multi-table database schema including University and College foreign keys.

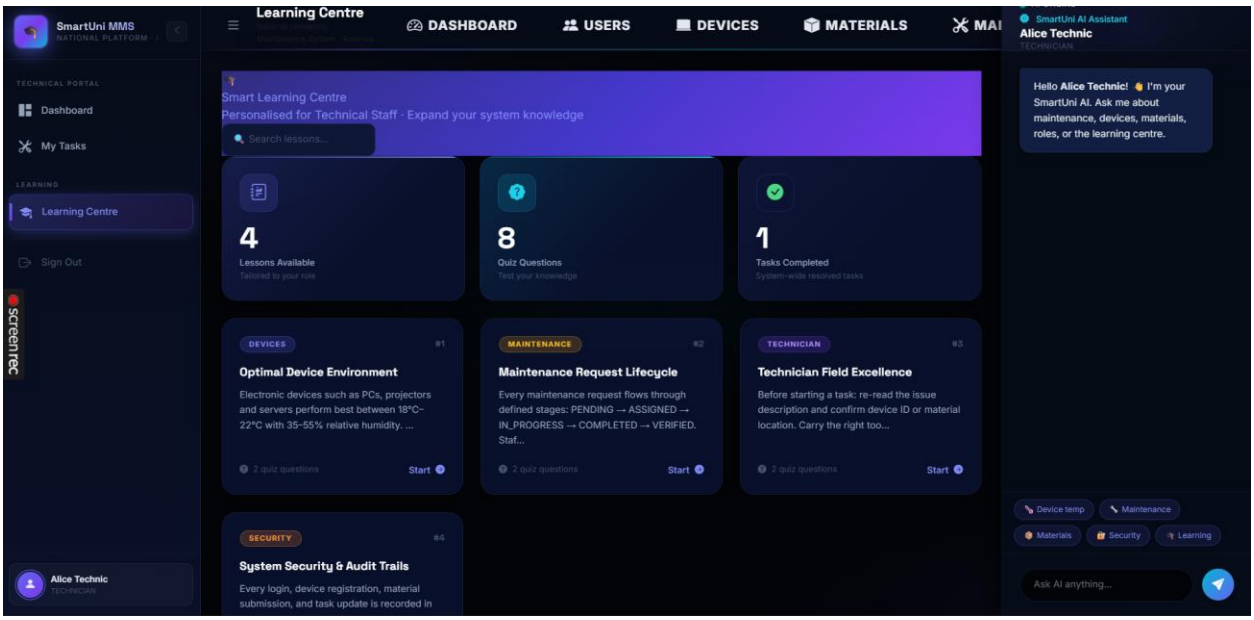
* **Integration Stress Test:** Confirmed that concurrent logins and state updates (Technician starting a task) are reconciled correctly in the Admin's real-time monitoring view.

Conclusion

Phase 9 successfully integrates the "Smart University Devices and Materials Maintenance System" into a unified platform. The connection between the advanced Java backend logic and the premium responsive frontend is fully verified, ensuring a robust and reliable system for national-scale deployment.




SmartUni MMS
NATIONAL PLATFORM
User Management
Dashboard
Users
Devices
Materials
ADMINISTRATION
Dashboard
Users
Devices
Materials
OPERATIONS
Maintenance
Analytics
LEARNING
Learning Centre
Sign Out
System Administrator
ADMIN
User Management
All Registered Users
Search...
NAME EMAIL ROLE UNIVERSITY STATUS LAST LOGIN ACTIONS
System Administrator admin@smartuni.rw ADMIN UR ACTIVE 19 Mar 16:12 Suspend
Alice Technic alicetechnic@smartuni.rw TECHNICIAN UR ACTIVE 14 Mar 19:45 Suspend
Bob Tech bobtech@smartuni.rw TECHNICIAN UR ACTIVE Never Suspend
Charlie Staff charliestaff@smartuni.rw STAFF UR ACTIVE 14 Mar 20:29 Suspend
Diana Staff dianastaff@smartuni.rw STAFF UR ACTIVE Never Suspend
Edward Cleaner edwardcleaner@smartuni.rw CLEANER_STUDENT UR ACTIVE 14 Mar 17:46 Suspend
Fionna Student fionastudent@smartuni.rw CLEANER_STUDENT UR ACTIVE Never Suspend
SmartUni AI Assistant
System Administrator
ADMIN
Hello System Administrator! I'm your SmartUni AI. Ask me about maintenance, devices, materials, roles, or the learning centre.
Device temp
Maintenance
Materials
Security
Learning
Ask AI anything...
screenrec


SmartUni MMS
NATIONAL PLATFORM
Learning Centre
DASHBOARD
USERS
DEVICES
MATERIALS
TECHNICAL PORTAL
Dashboard
My Tasks
LEARNING
Learning Centre
Sign Out
Alice Technic
TECHNICIAN
Smart Learning Centre
Personalised for Technical Staff · Expand your system knowledge
Search lessons...
4
Lessons Available
Tailored to your role
8
Quiz Questions
Test your knowledge
1
Tasks Completed
System-wide resolved tasks
DEVICES #1
Optimal Device Environment
Electronic devices such as PCs, projectors and servers perform best between 18°C–22°C with 35–55% relative humidity. ...
2 quiz questions
Start
MAINTENANCE #2
Maintenance Request Lifecycle
Every maintenance request flows through defined stages: PENDING → ASSIGNED → IN_PROGRESS → COMPLETED → VERIFIED. Staf...
2 quiz questions
Start
TECHNICIAN #3
Technician Field Excellence
Before starting a task: re-read the issue description and confirm device ID or material location. Carry the right too...
2 quiz questions
Start
SECURITY #4
System Security & Audit Trails
Every login, device registration, material submission, and task update is recorded in
SmartUni AI Assistant
Alice Technic
TECHNICIAN
Hello Alice Technic! I'm your SmartUni AI. Ask me about maintenance, devices, materials, roles, or the learning centre.
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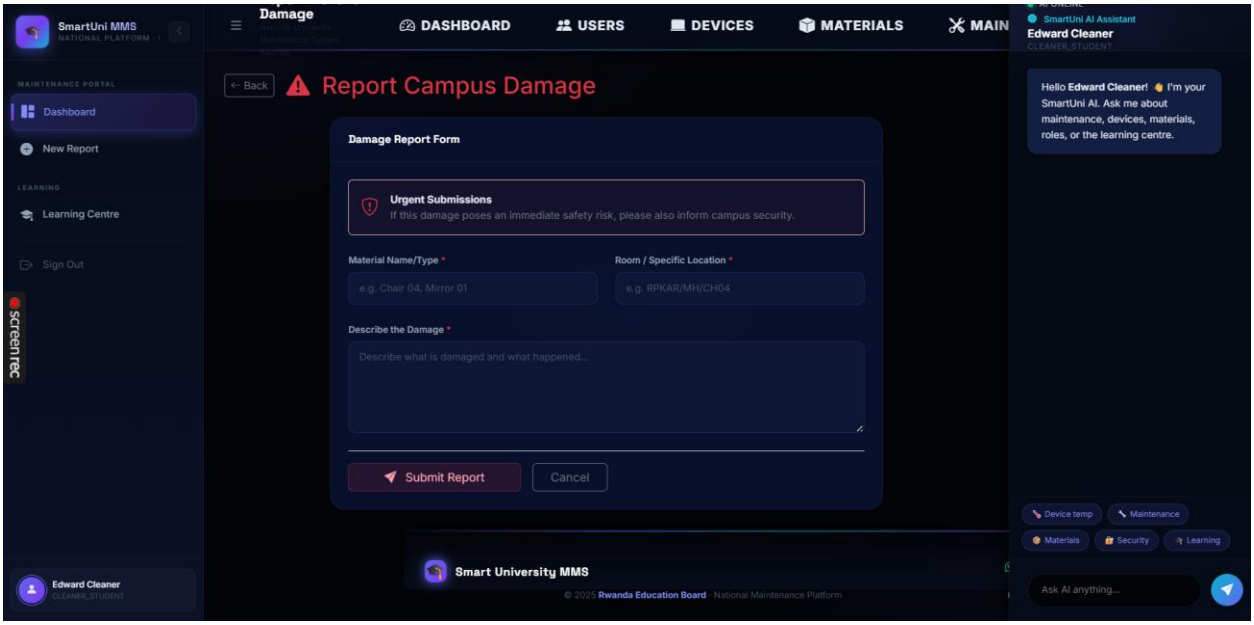
SmartUni MMS
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MAINTENANCE PORTAL
Dashboard
New Report
LEARNING
Learning Centre
Sign Out
Edward Cleaner
CLEANER_STUDENT
Damage
DASHBOARD
USERS
DEVICES
MATERIALS
MAIN
← Back
Report Campus Damage
Damage Report Form
Urgent Submissions
If this damage poses an immediate safety risk, please also inform campus security.
Material Name/Type *
e.g. Chair 04, Mirror 01
Room / Specific Location *
e.g. RPKAR/MH/CH04
Describe the Damage *
Describe what is damaged and what happened...
Submit Report
Cancel
Smart University MMS
© 2025 Rwanda Education Board · National Maintenance Platform
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